MILITARY SPECIFICATION

CALCULATOR SET, NUCLEAR, M28A1

1. SCOPE

1.1 This specification covers one type of nuclear calculator set consisting of one ABC-M1 radiac calculator and one M4A1 nuclear yield calculator in a plastic case.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

NN-P-530	-	Plywood, Flat-Panel.
TT-W-572	-	Wood-Preservative, Water-Repellent.
PPP-B-601	-	Boxes, Wood, Cleated-Plywood.
PPP-B-636	-	Box, Fiberboard.
PPP-C-843	-	Cushioning Material, Cellulosic.
PPP-F-320	-	Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes.

MILITARY

MIL-P-116	-	Preservation, Methods of.
MIL-P-15011	-	Pallets, Material Handling, Wood, Post Construction, 4-Way Entry.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:57Z
Check the source to verify that this is the current version before use.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-147 - Palletized and Containerized Unit Loads 40" x 48" Pallets, Skids, Runners, or Pallet-Type Base.

DRAWINGS

US ARMY MUNITIONS COMMAND

EDGEWOOD ARSENAL

DL5-86-34 - Calculator Set, Nuclear, M28A1

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification.

(Application for copies of these ratings, rules, and regulations should be addressed to Uniform Classification Committee, 202 Union Station, 516 West Jackson Boulevard, Chicago, Illinois 60606.)

3. REQUIREMENTS

3.1 Materials and components.

3.1.1 Materials. All materials cited on Drawing DL5-86-34 or on the subsidiary drawings shall conform to the specifications listed thereon, or to the specific characteristics set forth on the drawings.

3.1.2 Components. All components of the calculator set shall conform to the specifications and drawings listed on Drawing DL5-86-34 and subsidiary drawings.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:57Z
Check the source to verify that this is the current version before use.

3.2 Assembly. The calculators shall be placed into the case (Dwg C5-86-34).

3.3 Workmanship. The case shall be free from imperfections (pits, bumps, cracks, flash, abrasions, blisters, holes or porous areas); damage (rips, tears or punctures); and contamination (dirt, grease, or oil).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 Supplier's responsibility. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

4.1.2 Objective evidence. The supplier shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.2 Quality conformance inspection.

4.2.1 Lotting. A lot shall consist of the M28A1 nuclear calculator sets produced by one manufacturer, at one plant, from the same materials. under essentially the same manufacturing conditions. However, no more than one lot of ABC-M1 radiac calculators and one lot of M4A1 nuclear yield calculators shall be represented in any one lot of finished M28A1 nuclear calculator sets.

4.2.2 Sampling for examination. Sampling shall be conducted in accordance with MIL-STD-105.

4.2.3 Inspection procedure.

4.2.3.1 Examination. Sample calculator sets and the level A preparation for delivery shall be examined in accordance with the classification of defects and MIL-STD-105.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:57Z
Check the source to verify that this is the current version before use.

4.2.3.2 Classification of defects.

(a) Calculator set, nuclear, M28A1 (Dwg C5-86-34).

Categories	Defects
Critical:	None defined
Major:	AQL 2.5 percent defective
101	Component missing
102	Case dimensions
103	Case heat seal
104	Holding strip
105	Marking missing, incorrect or illegible
106	Workmanship (see 3.3)

(b) Preparation for delivery (section 5).

Categories	Defects
Critical:	None defined
Major:	AQL 6.5 percent defective
101	Packaging method
102	Filler pads
103	Container
104	Quantity per container
105	Container closure
106	Container bag damaged (broken, torn, crushed, or punctured)
107	Marking missing, incorrect or illegible.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A. The calculator set shall be individually packaged in accordance with method IC-3 of MIL-P-116. Prior to bag closure, a fiberboard filler the same size as the outside dimensions of the calculator case and conforming to type CF, grade 125 of PPP-F-320 shall be inserted in each bag to act as a stiffener.

5.1.2 Level C. The calculator set shall be packaged to provide adequate protection against deterioration and physical damage from the supply source to the first receiving activity for immediate use.

Source: https://assist.dla.mil -- Downloaded: 20261O-22T20:57Z
Check the source to verify that this is the currentversion before use.

5.2 Packing. Level shall be as specified (see 6.2).

5.2.1 Level A. One hundred calculators, packaged as specified in 5.1.1, shall be packed within an overseas type cleated-plywood box conforming to PPP-B-601. Cushioning material, conforming to type II, class A of PPP-C-843, shall be used to fill all voids within the box. The box shall be closed and strapped in accordance with the appendix of PPP-B-601. Galvanized strapping shall be used. The box or wood parts thereof shall be preserved by completely immersing in a water-repellent wood preservative conforming to type I or II, composition A, of TT-W-572, for a minimum of three minutes.

5.2.2 Level B. One hundred calculators, packaged as specified in 5.1.1, shall be packed in a V3c, style RSC fiberboard box conforming to PPP-B-636. In addition to box closure requirements, the box seams and manufacturer's joint shall be taped. Tape banding shall be used in lieu of metal banding as a box reinforcement.

5.2.3 Level C. One hundred calculator sets, packaged as specified in 5.1.2, shall be packed to insure carrier acceptance and safe delivery to the first domestic destination. Containers shall be in compliance with Uniform Freight Classification rules or the regulations of other common carriers applicable to the mode of transportation.

5.2.4 Palletization. The calculator sets, packed as specified in 5.2.1, shall be palletized in accordance with MIL-STD-147. Pallets shall conform to style 1, type I, class 1 of MIL-P-15011. The overall maximum height of the pallet load shall not exceed 54 inches, except in the event shipments are made by SEAVAN then the overall maximum height shall not exceed 43 inches. Wood treatment as specified in 5.2.1 shall be applied to the pallet.

5.3 Marking.

5.3.1 Unit and shipping container. In addition to any special marking required by the contract or order, unit packages shall be marked in accordance with MIL-STD-129. Shipping containers shall be marked in accordance with Drawing D150-2-1.

5.3.2 Pallet loads. A plywood marking panel measuring 48 by 24 by 1/4 inches shall be used to reflect the appropriate marking of the pallet load. The marking panel shall be marked in accordance with MIL-STD-129. The plywood shall be weather resistant, container grade, conforming to NN-P-530.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:57Z
Check the source to verify that this is the current version before use.

6. NOTES

6.1 Intended use. This calculator set is intended for use in radiological warfare for calculating past, present or future dose rates and nuclear yield.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Level of packaging and packing required.

Custodians:

Army - MU

Preparing activity:

Army - MU(EA)

Project No. 6665-A237

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:57Z
Check the source to verify that this is the current version before use.

TO DETACH FORM - CUT ON THIS LINE

SPECIFICATION ANALYSIS SHEET

Form Approved Budget Bureau No. 119 R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:57Z
Check the source to verify that this is the current version before use.